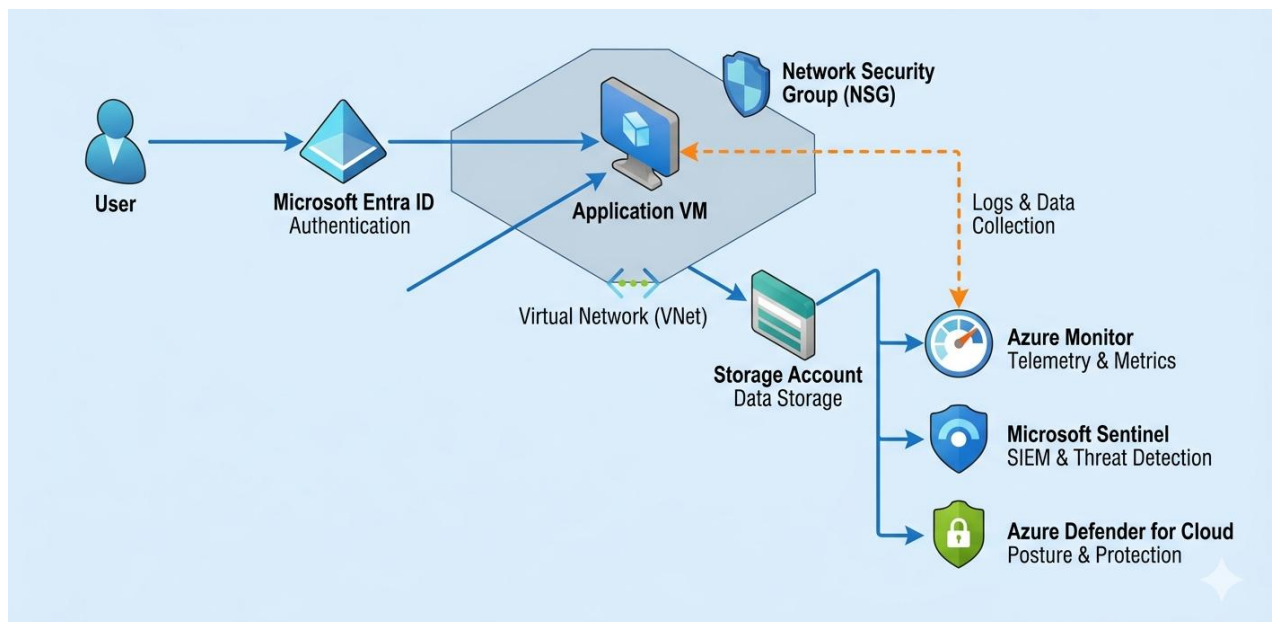


Project Title: Secure Azure Cloud Environment with Monitoring & Threat Detection

Objective: The objective of this project is to design and implement a secure cloud environment in Microsoft Azure by applying best practices in:

- Identity & Access Management
- Network security
- Secure storage
- Logging & monitoring
- Threat Detection and SIEM Integration

Project Architecture



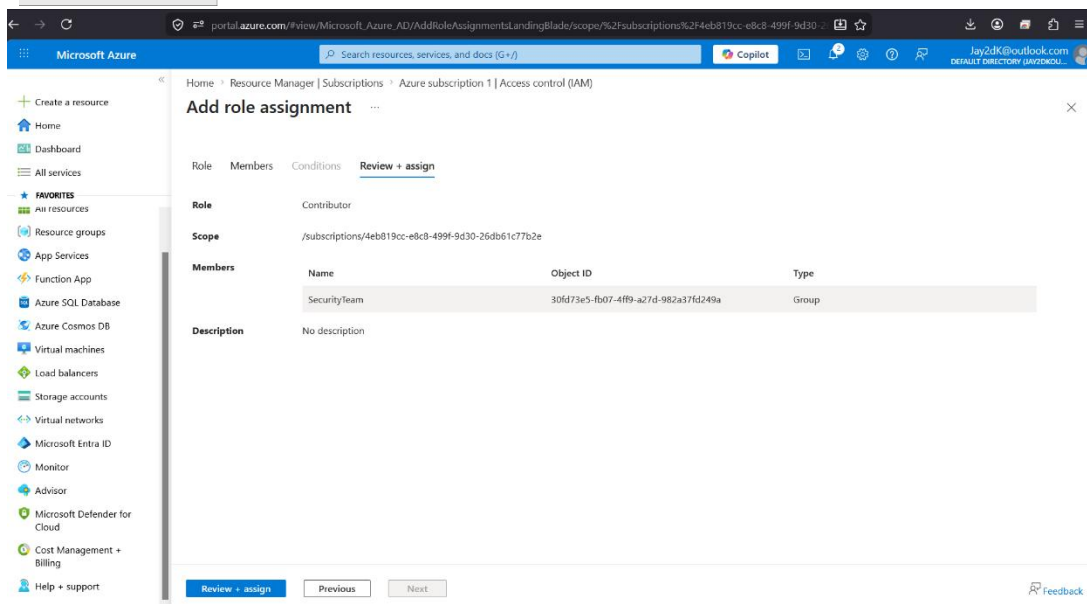
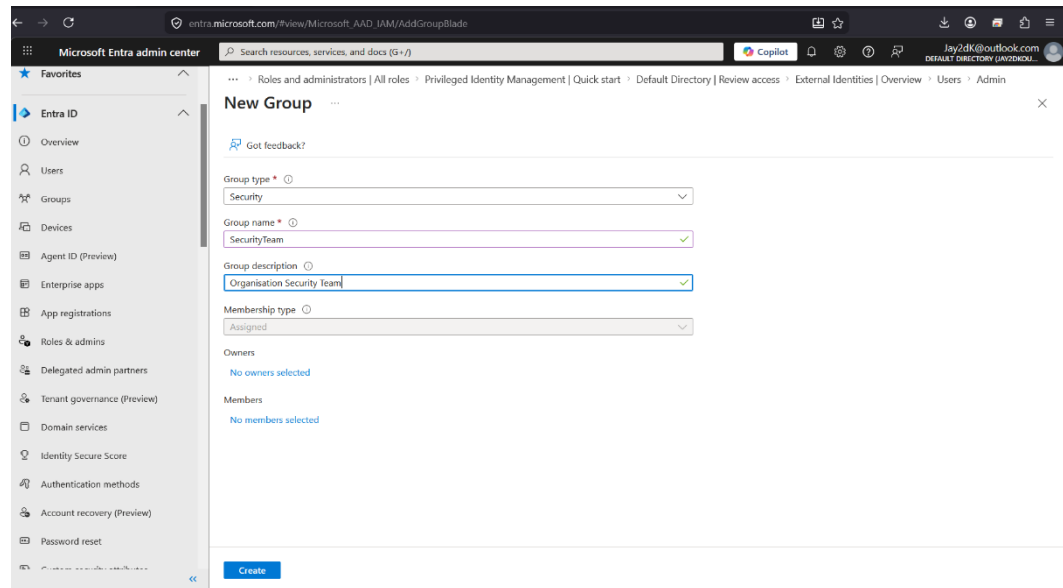
1. Account Foundation & MFA

- Sign Up: Create an Azure Free Account.
- MFA Activation: Go to Microsoft Entra ID > Security > Multifactor authentication.
- Action: Enable MFA for your primary administrative account to prevent credential-based attacks.

2. Identity & Access Management (RBAC)

- User Creation: In Entra ID, create a new user (*Admin*).
- Group Management: Create a group named *SecurityTeam* and add the new user.

- **RBAC Assignment:** Go to your Subscription > Access Control (IAM). Assign the Contributor role to the SecurityTeam group.



3. Secure Network Architecture (VNet)

- **VNet Creation:** Go to Virtual Networks → Create
Name: *VNet1*, Address space: *10.0.0.0/25*
Create subnets:
- Public: *10.0.0.0/28*
- Private: *10.0.0.64/26*
- **NSG Configuration:** Create a Network Security Group(*VnetNSG1*). Add an inbound rule for Port 22 (SSH) but change the Source to My IP Addresses
- Then associate the NSG to the public subnet

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

Jay2dK@outlook.com

DEFAULT DIRECTORY (JAY2DKOU...)

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > Network foundation | Virtual networks

Create virtual network

Validation passed

Basics Security Address space Tags Review + create

Basics

Subscription

Azure subscription 1

Resource Group

SecurityProject1

Name

VNet1

Region

South Africa North

Security

Azure Bastion

Disabled

Azure Firewall

Disabled

Azure DDoS Network Protection

Disabled

Address space

Address space

10.0.0.0/25 (128 addresses)

Subnet

Public_Subnet (10.0.0.0/28) (16 addresses)

Subnet

Private_Subnet (10.0.0.64/26) (64 addresses)

- Private subnet

Enabled

Previous

Next

Create

Download a template for automation

Give feedback

Microsoft Azure

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Microsoft Defender for Cloud

Cost Management + Billing

Home > Network foundation | Network security groups

Create network security group

Validation passed

Basics Tags Review + create

Basics

Subscription

Azure subscription 1

Resource group

SecurityProject1

Region

South Africa North

name

VNetNSG1

Tags

None

Create

< Previous

Next >

Download a template for automation

Microsoft Azure portal interface showing the 'VnetNSG | Subnets' page. The left sidebar contains navigation options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'VnetNSG | Subnets' page with a search bar and a list of subnets. A modal window titled 'Associate subnet' is open, showing options to select a virtual network (VNet) and a subnet. A notification banner at the top right indicates 'Saving subnet'.

Home > VnetNSG

VnetNSG | Subnets

Network security group

Search

+ Associate

Search subnets

Name	Address range
No results.	

Virtual network: VNet (SecurityProject)

Subnet: Public_Subnet

OK

*** Saving subnet

Saving network security group for subnet 'Public_Subnet'.

Microsoft Azure portal interface showing the 'VnetNSG | Inbound security rules' page. The left sidebar contains navigation options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'VnetNSG | Inbound security rules' page with a search bar and a list of security rules. A modal window titled 'Add inbound security rule' is open, showing fields for source, destination, service, and action.

Home > CreateNetworkSecurityGroupBladeV2-20260416141938 | Overview > VnetNSG

VnetNSG | Inbound security rules

Network security group

Search

+ Add

Hide default rules

Refresh

Delete

Filter by name

Port == all

Priority	Name	Port
65000	AllowVnetInBound	Any
65001	AllowAzureLoadBalanc...	Any
65500	DenyAllInBound	Any

Network security group security rules are evaluated by priority using deny the traffic. A security rule can't have the same priority and direction that have a higher priority. [Learn more](#)

Source: My IP address

Source IP addresses/CIDR ranges: 102.218

Source port ranges: *

Destination: Any

Service: SSH

Destination port ranges: 22

Protocol: TCP

Action: Allow

Add Cancel

Give feedback

4. Virtual Machine Hardening

- Deployment: Create a Linux VM (AdminJK) in the Public Subnet.
- Disable Passwords: Edit `/etc/ssh/sshd_config`, set *PasswordAuthentication no* and *PermitRootLogin no*
- Update: Run `sudo apt update && sudo apt upgrade -y` to patch vulnerabilities.

The screenshot shows the 'Create a virtual machine' page in the Azure portal. The 'Validation passed' status is displayed at the top. The 'Disks' section is expanded, showing the following configuration:

Property	Value
OS disk size	Image default
OS disk type	Premium SSD LRS
Use managed disks	Yes
Delete OS disk with VM	Enabled
Ephemeral OS disk	No

The 'Networking' section is also expanded, showing the following configuration:

Property	Value
Virtual network	VNet1
Subnet	Public_Subnet (10.0.0.0/28)
Public IP	(new) LinuxVM1-ip
NIC network security group	None
Accelerated networking	On
Place this virtual machine behind an existing load balancing solution?	No
Delete public IP and NIC when VM is deleted	Disabled

The 'Management' section shows 'Microsoft Defender for Cloud' set to 'Basic (free)'. Navigation buttons include '< Previous', 'Next >', and 'Create'. A link to 'Download a template for automation' and a 'Give feedback' button are also present.

The screenshot shows the 'Create a virtual machine' page in the Azure portal, with the 'Basics' section expanded. The configuration is as follows:

Property	Value
Subscription	Azure subscription 1
Resource group	SecurityProject1
Virtual machine name	LinuxVM1
Region	South Africa North
Availability options	No infrastructure redundancy required
Zone options	Self-selected zone
Security type	Standard
Image	Ubuntu Server 24.04 LTS - Gen2
VM architecture	x64
Size	Standard B2ats v2 (2 vcpus, 1 GiB memory)
Enable Hibernation	No
Authentication type	SSH public key
Username	AdminJK
SSH Key format	RSA
Key pair name	LinuxVM1_key
Azure Spot	No

The 'Disks' section is partially visible at the bottom. Navigation buttons include '< Previous', 'Next >', and 'Create'. A link to 'Download a template for automation' and a 'Give feedback' button are also present.

```
AdminJK@LinuxVM1: /etc
GNU nano 7.2 /etc/ssh/sshd_config
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

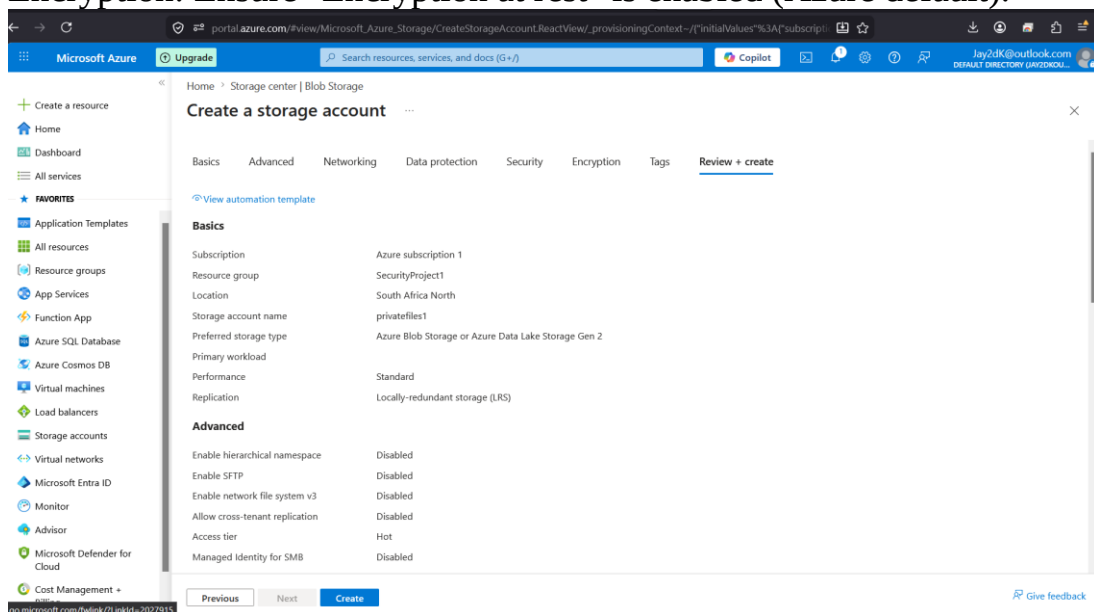
# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication no
#PermitEmptyPasswords no

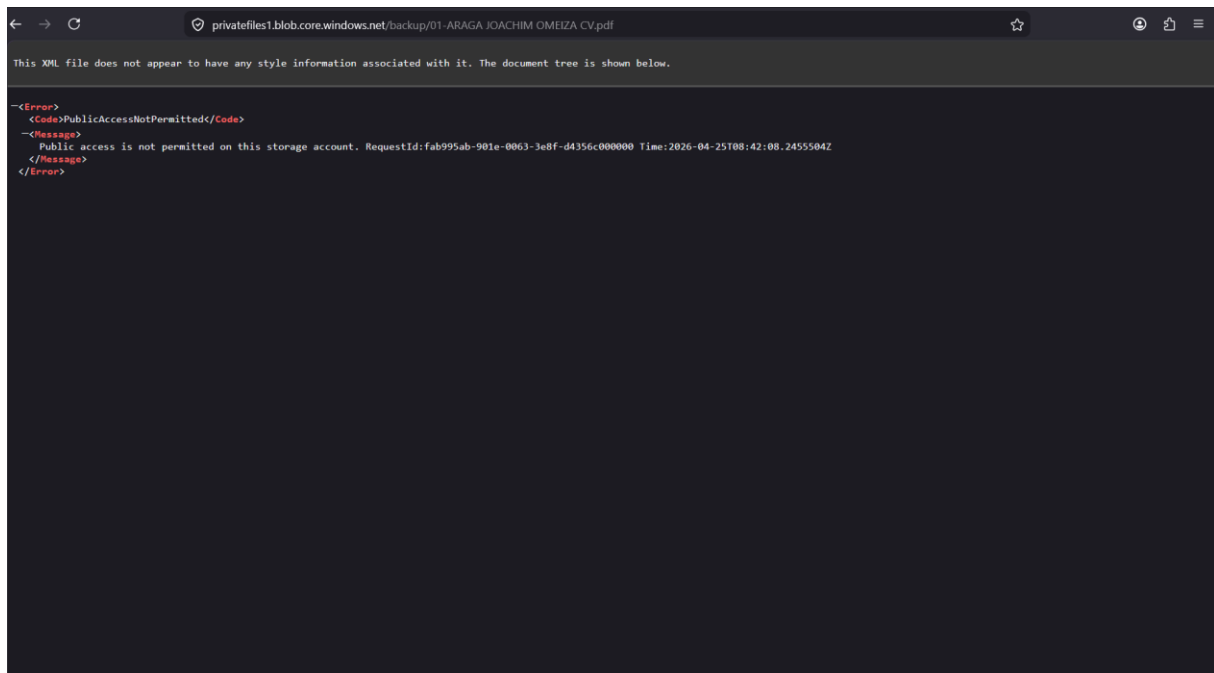
# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark
Exit Read File Replace AU Paste AJ Justify AV Go To Line M-E Redo M-6 Copy
```

5. Secure Storage Configuration

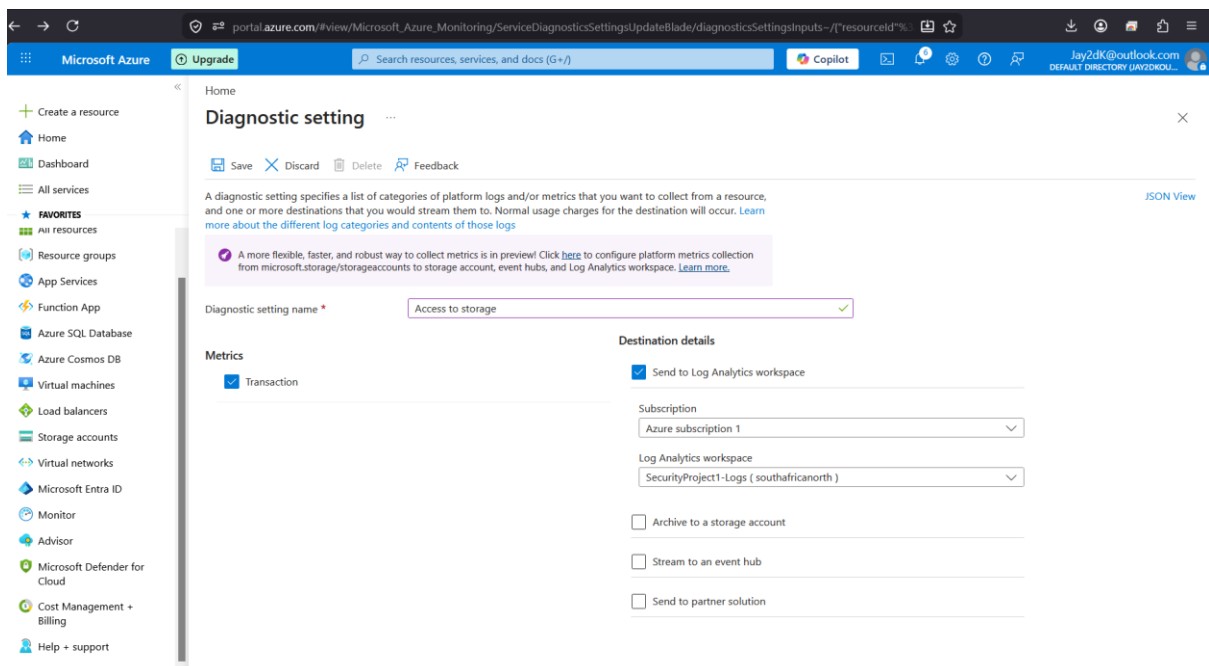
- Storage Account: Create a new storage account.
- Hardening: Go to Configuration > Set Allow storage account key access to Disabled and enforce Minimum TLS version 1.2.
- Encryption: Ensure "Encryption at rest" is enabled (Azure default).





6. Logging & Monitoring Setup

- Log Analytics: Create a Log Analytics Workspace (LAW).
- Diagnostics: Go to your VM and Storage Account > Diagnostic settings > Add diagnostic setting. Send all logs to your LAW.
- Configure Data Collection Rule (DCR)
- Enable Syslog (auth, authpriv)
- Verify logs in workspace



portal.azure.com/#view/Microsoft_OperationsManagementSuite_Workspace/CreateWorkspaceBladeV2/_provisioningContext-~/InitialVal...

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Log Analytics workspaces

Create Log Analytics workspace

Validation passed

Basics Tags Review + Create

Log Analytics workspace
by Microsoft

Basics

Subscription	Azure subscription 1
Resource group	SecurityProject1
Name	SecurityProject1-Logs
Region	South Africa North

Pricing

Pricing tier Pay-as-you-go (Per GB 2018)

The cost of your workspace depends on the volume of data ingested and how long it is retained. Regional pricing details are available on the [Azure Monitor pricing page](#). You can change to a different pricing tier after the workspace is created. [Learn more](#) about Log Analytics pricing models.

Tags

None

Create Previous Download a template for automation

portal.azure.com/#view/Microsoft_Azure_Monitoring/DcrConfigureDataflow.ReactView/isNewDcr-~/true/isOpenedByKnockoutBlade-~/false

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Monitor | Data Collection Rules > Create Data Collection Rule

Add new data source

Data source Destination

Select which data source type and the data to collect for your resource(s).

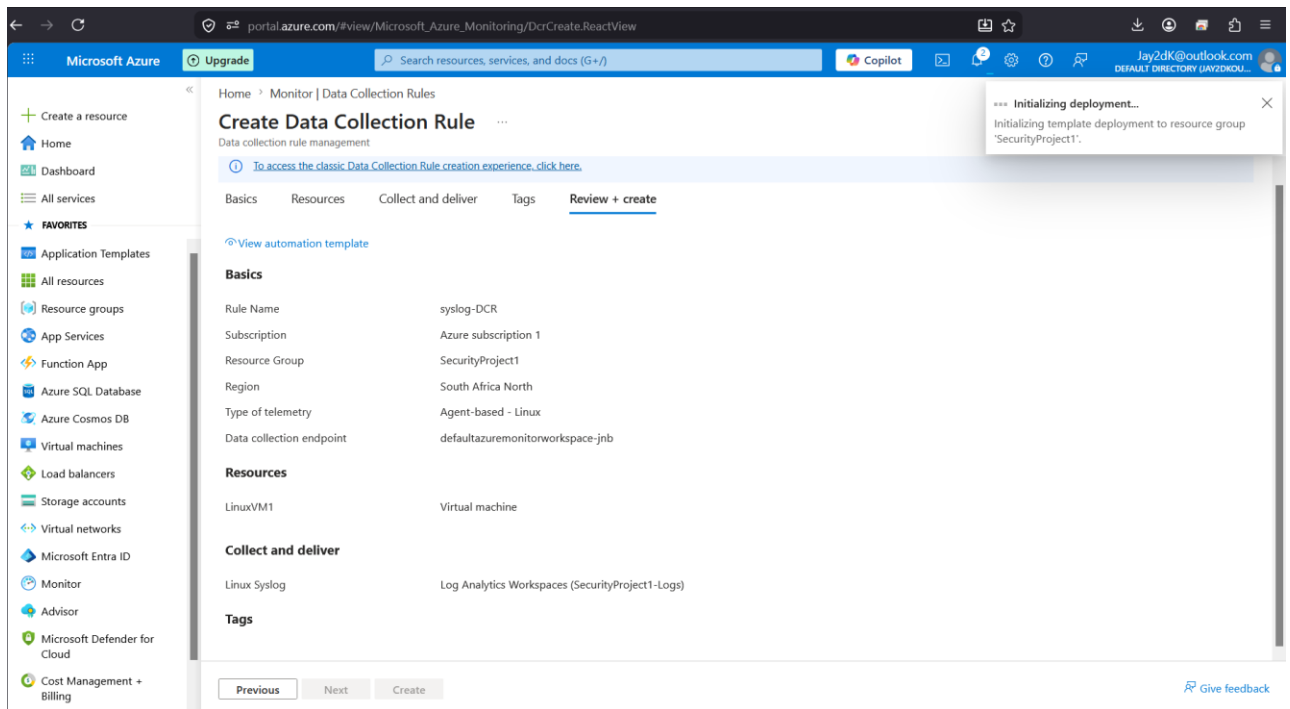
Data source type * Linux Syslog

Set minimum log level for selected facilities: None

Facility	Log Level
☐ LOG_ALERT	None
☐ LOG_AUDIT	None
☐ LOG_AUTH	LOG_DEBUG
☐ LOG_AUTHPRIV	LOG_DEBUG
☐ LOG_CLOCK	None
☐ LOG_CRON	None
☐ LOG_DAEMON	None
☐ LOG_FTP	None

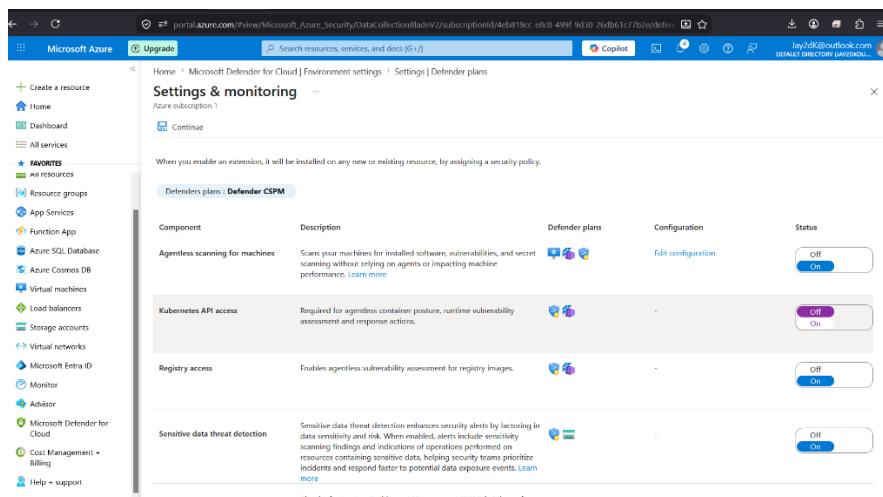
Save Next: Destinations > Cancel

Feedback



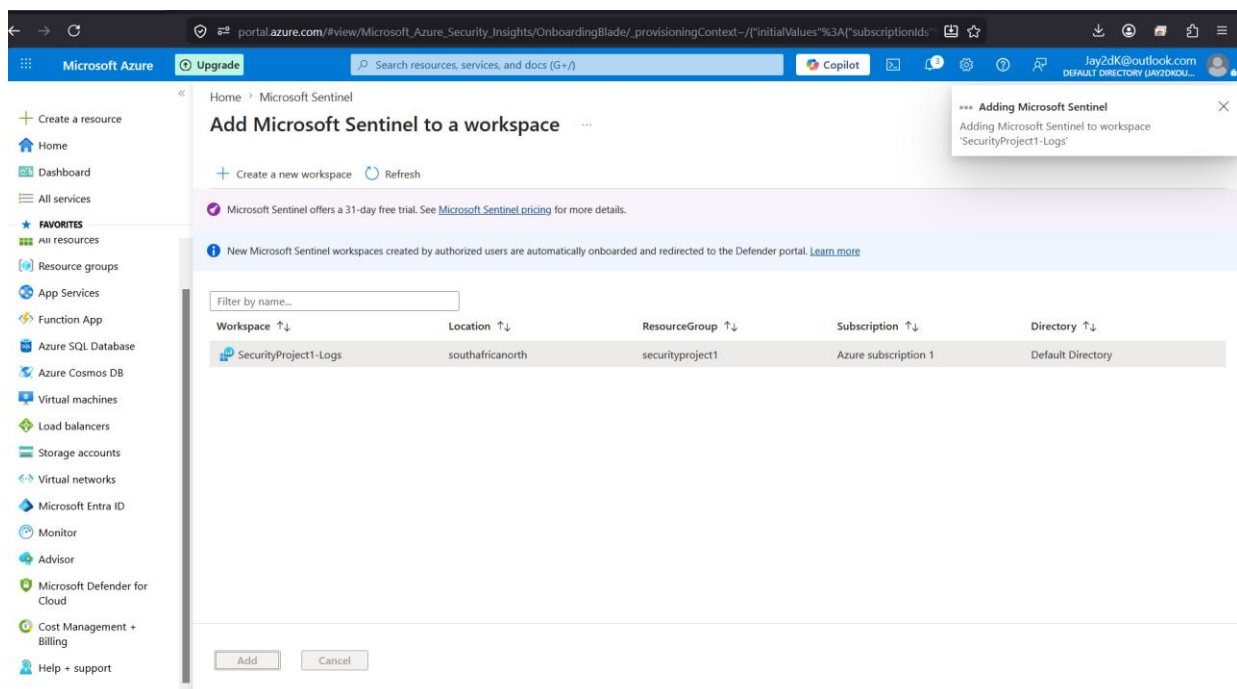
7. Posture Management (Defender)

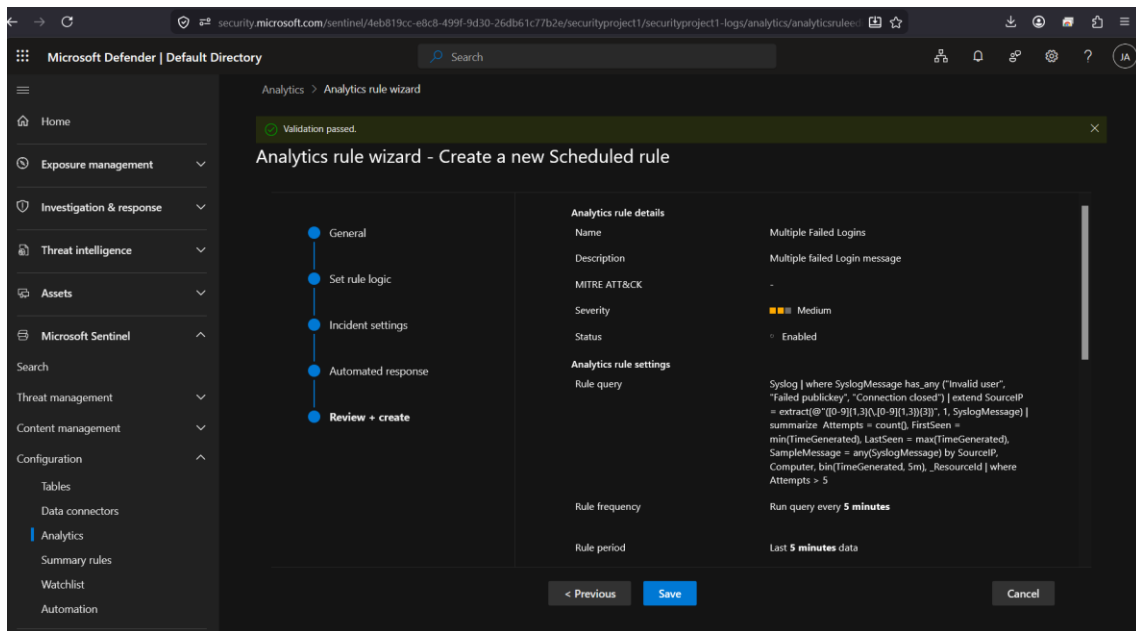
- Activation: Open Microsoft Defender for Cloud and click Upgrade to enable enhanced security features.
- Review: Check your Secure Score and remediate any "High" priority recommendations, such as open ports or missing updates.



8. SIEM Integration (Microsoft Sentinel)

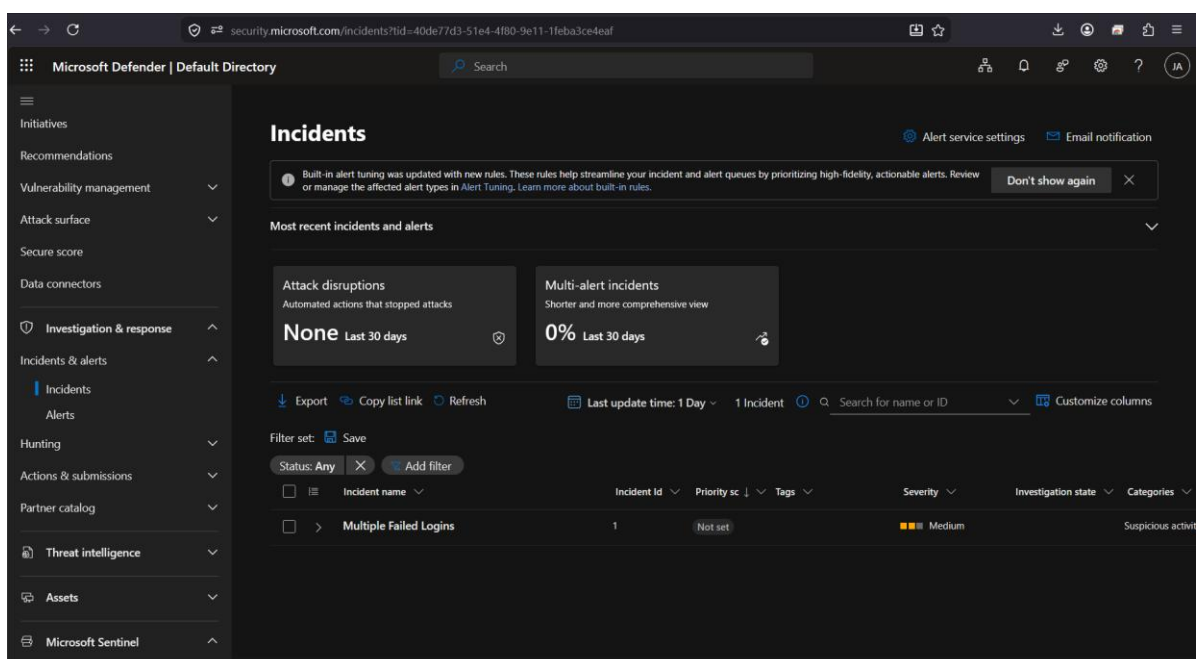
- Onboarding: Search for Microsoft Sentinel > Create > Select your existing LAW.
- Connectors: Install the Azure Activity and Microsoft Defender for Cloud data connectors.
- Detection Rule: In Sentinel, go to Analytics > Scheduled query rule. Use this KQL logic: *Syslog | where SyslogMessage has_any ("Invalid user", "Failed publickey", "Connection closed") | extend SourceIP = extract(@"([0-9]{1,3}\.){1,3}([0-9]{1,3}){3})", 1, SyslogMessage) | summarize Attempts = count(), FirstSeen = min(TimeGenerated), LastSeen = max(TimeGenerated), SampleMessage = any(SyslogMessage) by SourceIP, Computer, bin(TimeGenerated, 5m), _ResourceId | where Attempts > 5*





9. Incident Response Simulation

- The Attack: Attempt to log into your VM from an unauthorized IP multiple times.
- The Detection: Monitor the Incidents blade in Sentinel for the "Multiple Failed Logins" alert.
- The Response: Use the log data to identify the attacker's IP and add a Deny rule for that IP in your NSG.



Microsoft Azure portal showing the Logs view for a Security Project. The interface includes a sidebar with navigation options like Home, Dashboard, All services, and Favorites. The main content area displays a table of logs with columns: SourceIP, Computer, TimeGenerated [UTC], Attempts, FirstSeen [UTC], LastSeen [UTC], and SampleMessage. The logs show activity from a LinuxVM1 on 5/3/2026, including an invalid user login attempt and a connection closure.

SourceIP	Computer	TimeGenerated [UTC]	Attempts	FirstSeen [UTC]	LastSeen [UTC]	SampleMessage
102.218.200.22	LinuxVM1	5/3/2026, 8:25:00.000 PM	9	2026-05-03T20:25:55.45204Z	2026-05-03T20:29:59.589748Z	Connection closed by authenticating user AdminJK 102.218.200.22 port 50725 [preauth]
102.218.200.22	LinuxVM1	5/3/2026, 8:25:00.000 PM	9	2026-05-03T20:25:55.45204Z	2026-05-03T20:29:59.589748Z	Invalid user user from 102.218.200.22 port 50727

Microsoft Azure portal showing the Logs view for a Security Project, specifically the 'Create an alert rule' interface. The interface includes a sidebar with navigation options like Home, Monitor, and Alerts. The main content area displays a table of logs with columns: SourceIP, Computer, TimeGenerated [UTC], Attempts, SampleMessage, and FirstSeen [UTC]. The logs show activity from a LinuxVM1 on 5/3/2026, including an invalid user login attempt and a connection closure.

SourceIP	Computer	TimeGenerated [UTC]	Attempts	SampleMessage	FirstSeen [UTC]
102.218.200.22	LinuxVM1	5/3/2026, 9:30:00.000 PM	30	Invalid user user from 102.218...	5/3/2026, 9:31:20.732 PM
102.218.200.22	LinuxVM1	5/3/2026, 8:25:00.000 PM	9	Connection closed by authentic...	5/3/2026, 8:25:55.452 PM

Microsoft Azure portal interface showing the configuration of a Network Security Group (NSG) named "blocked_IP". The page displays the "Inbound security rules" section, which lists several rules. The first rule, "blocked_ip", is highlighted and shows a port of 8080, protocol of Any, source of 102.218, and action of Deny. Other rules include "AllowVnetInBound", "AllowAzureLoadBalan...", and "DenyAllInBound".

Home > Network foundation | Network security groups > blocked_IP

blocked_IP | Inbound security rules

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name: Port == all Protocol == all Source == all Destination == all Action == all

Name	Port	Protocol	Source	Destination	Action
blocked_ip	8080	Any	102.218	Any	Deny
AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	Allow
DenyAllInBound	Any	Any	Any	Any	Deny

Left sidebar navigation menu:

- Create a resource
- Home
- Dashboard
- All services
- FAVORITES
- Application Templates
- All resources
- Resource groups
- App Services
- Function App
- Azure SQL Database
- Azure Cosmos DB
- Virtual machines
- Load balancers
- Storage accounts
- Virtual networks
- Microsoft Entra ID
- Monitor
- Advisor
- Microsoft Defender for Cloud
- Cost Management + Billing

Right sidebar navigation menu:

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Inbound security rules
 - Outbound security rules
- Network interfaces
 - Subnets
 - Properties
 - Locks
- Monitoring
- Automation
- Help

Footer: Add or remove favorites by pressing Ctrl+Shift+F

Project by: Joachim Araga